

WEXFORD COUNTY AP, MI, USA

WMO: 726384

Lat: 44.283N

Lon: 85.417W

Elev: 398

StdP: 96.64

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-18.0	-25.0	0.4	-21.1	-21.4	0.6	-17.6	10.4	-4.9	9.3	-4.8	1.0	250	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	30.9	21.7	28.8	20.4	27.5	19.7	23.5	28.0	22.5	26.7	21.4	25.2	4.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.9	26.0	21.2	16.6	24.7	20.1	15.6	23.8	72.3	28.1	68.1	26.3	64.1	25.0	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.3	7.5	-27.4	33.5	4.7	1.9	-30.8	34.9	-33.6	36.0	-36.3	37.1	-39.7	38.5		
(5)			-27.2	25.4	4.2	1.8	-30.2	26.6	-32.7	27.7	-35.1	28.7	-38.2	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-7.0	-6.4	-1.5	5.5	12.1	17.4	19.7	18.7	14.8	8.3	1.8	-3.6		
	DBStd	10.75	6.07	6.29	6.54	5.27	4.94	4.08	3.56	3.31	4.53	4.83	4.99	5.28		
	HDD10.0	2313	528	458	361	154	33	2	0	0	9	94	251	423		
	HDD18.3	4423	786	692	614	386	202	64	27	36	123	314	497	681		
	CDD10.0	1112	0	0	5	19	99	223	300	268	153	41	3	0		
	CDD18.3	181	0	0	0	1	9	37	68	47	18	2	0	0		
	CDH23.3	2005	0	0	4	12	138	440	754	460	181	16	0	0		
	CDH26.7	493	0	0	1	1	28	118	210	98	37	1	0	0		
(14) Wind	WSAvg	3.2	3.6	3.6	3.4	3.7	3.4	2.9	2.6	2.5	2.7	3.3	3.5	3.5		
(15) Precipitation	PrecAvg	865	53	39	47	77	84	90	82	84	79	88	63	54		
	PrecMax	1049	102	82	104	120	172	247	233	185	155	233	138	134		
	PrecMin	672	17	15	16	24	14	34	34	35	21	21	26	20		
	PrecStd	111	24	17	23	29	42	50	48	36	30	48	30	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	11.3	21.9	25.1	29.6	32.5	33.3	32.2	30.5	25.9	17.8	12.5		
		MCWB	7.1	8.4	15.8	15.2	19.7	22.7	23.7	22.0	21.0	18.0	12.6	10.6		
	2%	DB	5.1	7.2	15.2	21.4	27.1	29.6	31.2	29.0	27.4	22.4	14.1	7.8		
		MCWB	3.4	4.4	9.6	13.0	18.2	20.3	22.3	20.8	19.1	16.5	11.1	6.2		
	5%	DB	2.5	4.0	11.5	18.0	24.7	27.7	28.7	27.5	25.4	19.7	12.4	5.1		
		MCWB	1.7	1.9	7.9	10.4	16.5	19.5	20.7	20.0	18.9	15.5	9.6	3.7		
	10%	DB	1.2	2.1	7.8	15.2	22.2	26.0	27.4	26.1	23.2	16.9	9.9	2.8		
		MCWB	0.6	0.8	4.6	9.3	15.2	18.6	20.0	19.2	17.8	13.3	7.5	1.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	8.4	16.2	17.0	21.7	23.9	25.8	24.4	23.0	20.6	13.9	10.8		
		MCDB	8.2	9.6	21.1	22.5	26.7	29.8	31.3	27.7	27.3	23.6	16.2	12.1		
	2%	WB	3.8	4.6	11.9	14.3	19.7	22.2	24.0	23.0	21.4	17.8	11.8	6.6		
		MCDB	4.9	6.9	14.2	19.5	24.4	27.4	28.5	26.9	24.4	20.7	14.0	7.8		
	5%	WB	1.8	2.3	7.9	12.2	18.0	21.1	22.8	21.9	20.2	15.7	9.8	3.8		
		MCDB	2.4	3.8	10.5	16.7	22.7	25.3	26.9	25.6	23.6	19.0	12.1	5.0		
	10%	WB	0.5	0.7	4.8	9.9	16.2	20.0	21.6	20.8	18.9	13.6	7.2	1.8		
		MCDB	1.2	2.0	7.8	14.7	20.5	24.1	25.4	23.8	22.8	16.4	9.4	2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	9.1	10.8	12.3	13.5	13.1	13.4	12.8	12.7	10.5	7.8	6.5		
		MCDBR	7.8	9.7	14.2	17.4	16.6	15.4	15.1	14.6	14.6	14.7	12.3	8.9		
	5% WB	MCWBR	7.0	7.8	9.7	10.4	9.0	7.9	7.4	7.6	8.1	9.3	9.3	7.6		
		MCWBR	7.3	8.7	12.9	14.8	13.4	12.2	12.2	11.4	11.4	12.7	11.2	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.291	0.315	0.343	0.372	0.391	0.388	0.377	0.361	0.331	0.297	0.273		
	taud		2.358	2.303	2.405	2.372	2.347	2.320	2.333	2.362	2.405	2.490	2.518	2.420		
	Ebn at Noon		854	899	924	920	898	878	876	874	862	847	827	821		
	Edh at Noon		77	98	102	115	121	125	122	115	101	82	67	66		
(44) All-Sky Solar Radiation	RadAvg		1.24	2.15	3.56	4.52	5.46	6.00	6.06	5.16	4.08	2.44	1.39	0.94		
	RadStd		0.11	0.23	0.41	0.43	0.50	0.40	0.31	0.33	0.39	0.30	0.20	0.11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page